

Course on Probabilistic Design of Coastal Structures

Statistical Modelling of Environmental Conditions

AWT- AUTOREGRESSIVE LOGISTIC REGRESSION MODEL

Jared Ortiz-Angulo Cantos

Researcher, GeoOcean Group, Universidad de Cantabria (UC)

ProCoast 2025
October 20th-23th, 2025



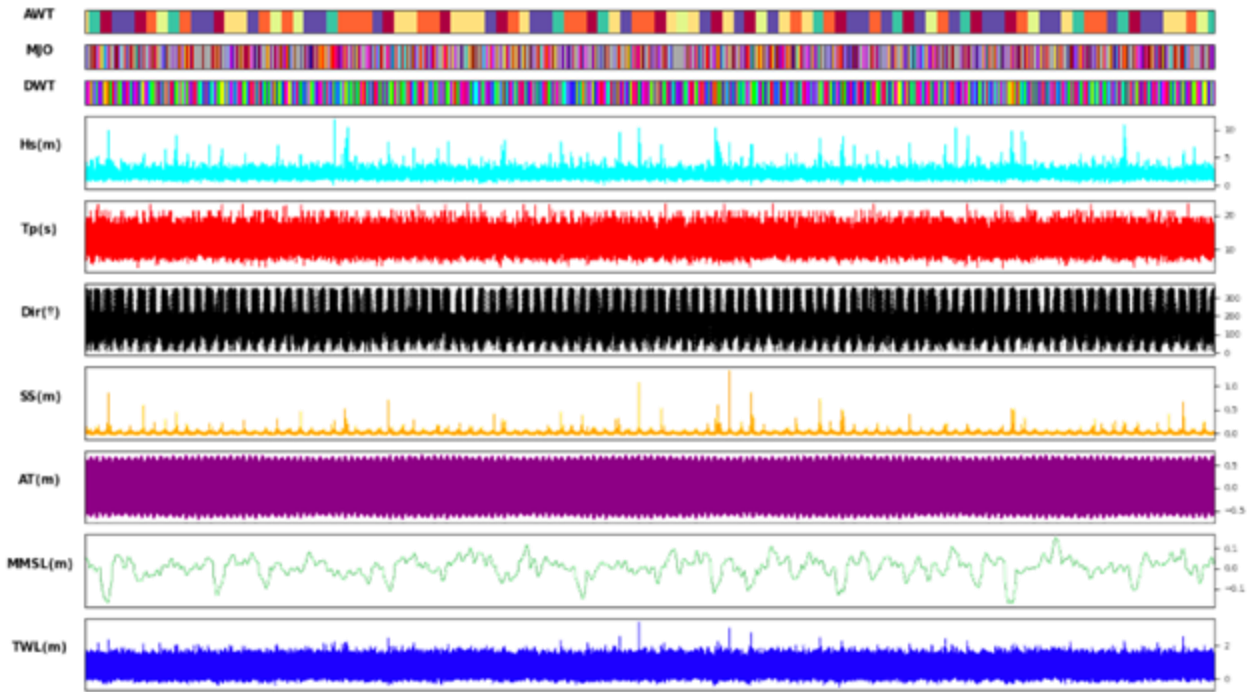
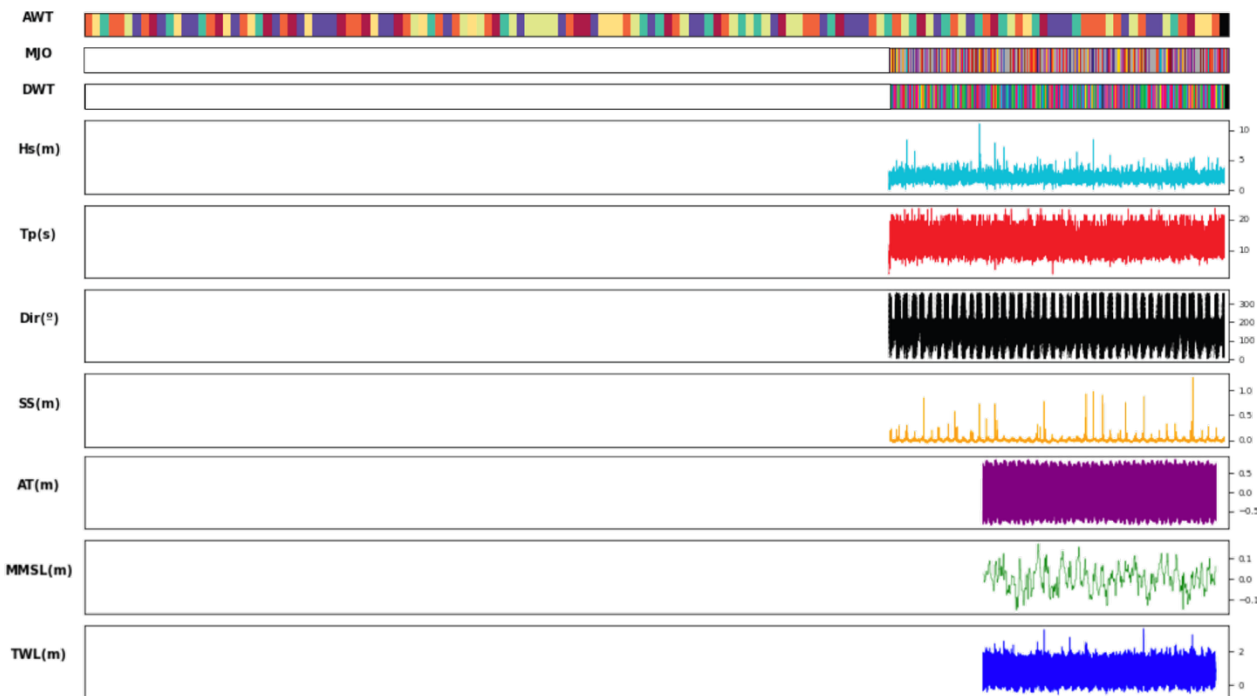
Introduction

TESLA

Anderson et al 2019

Cagigal et al 2020

N Simulations x n Years

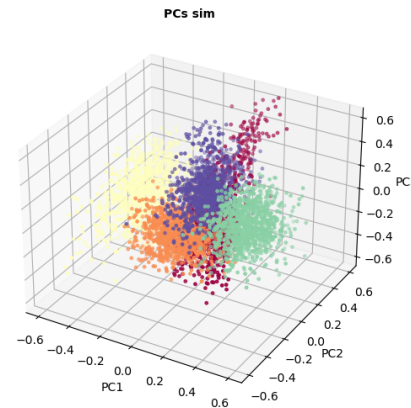
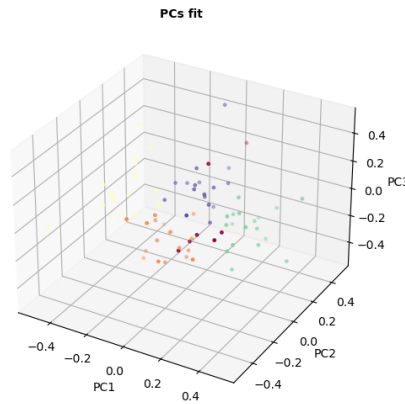


Framework

AWTs

1. PCs1,2,3 Random Generation with Copulas

Nº sims = 100



2. Autoregressive Logistic Regression – fit & simulation

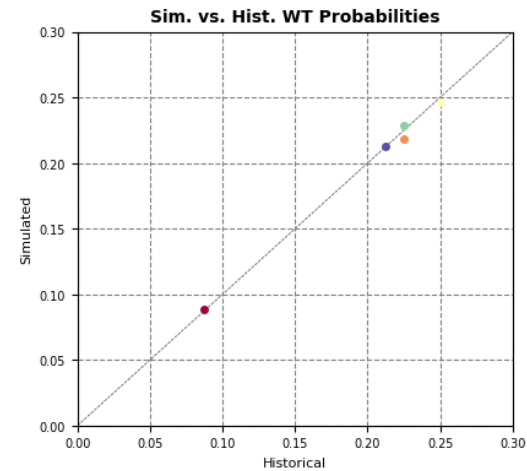
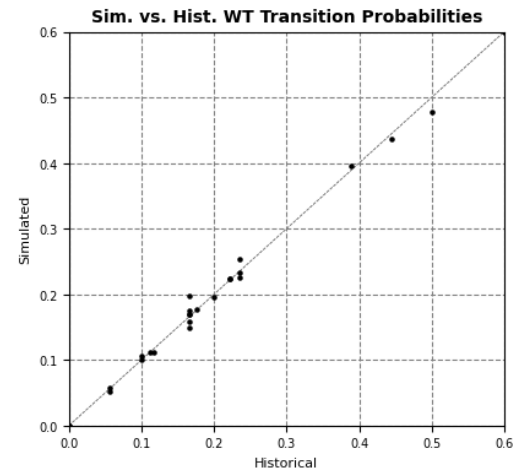
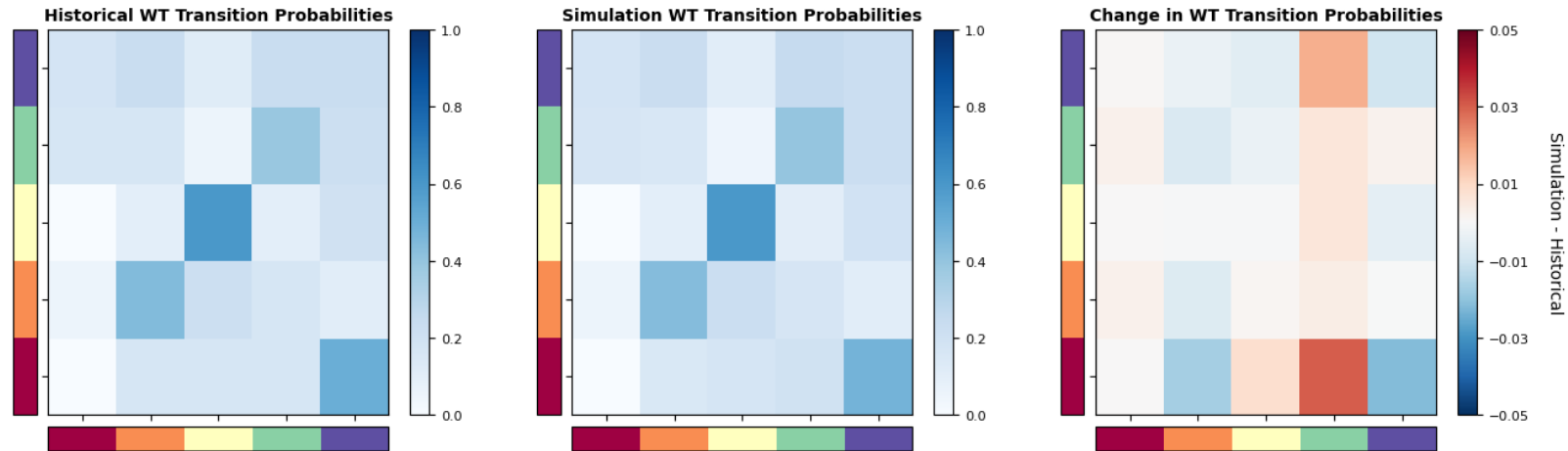
year ini = 1900
year end = 2000

MARKOV CHAIN
ORDER 1



Validation

Cluster Probabilities Transitions: All Simulations



Thank you

jared.ortizangulo@unican.es



<https://geocean.sci.unican.es/hyweb/>



GeoOcean



@GeoOcean_UC

ProCoast 2025
October 20th-23th, 2025

